LAB EXPERIMENT RECORD

Set 18: Product Inventory Management

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in $)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

1. Bar Chart

Aim

To create a bar chart to visualize the quantity of each product available in the inventory and label the axes and title the chart.

R Code

```
product_name <- c("Product A", "Product B", "Product C")

quantity <- c(250, 175, 300)

barplot(quantity,

    names.arg = product_name,

    main = "Quantity of Products in Inventory",
```

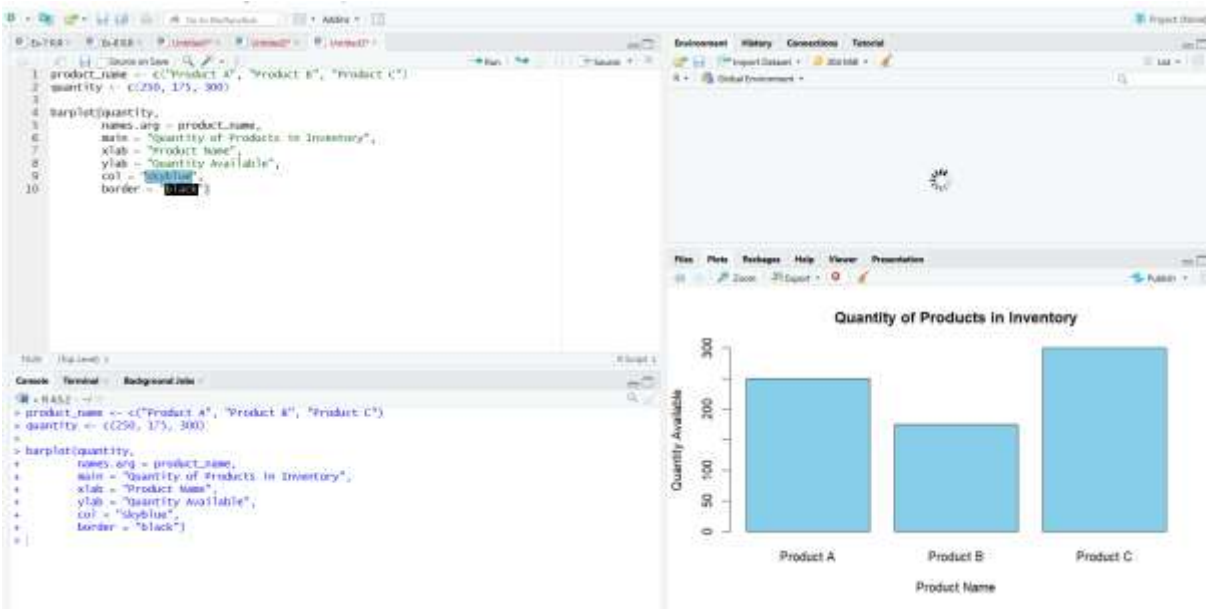
xlab = "Product Name",

ylab = "Quantity Available",

col = "skyblue",

border = "black")

Output



A bar chart showing the available quantities of Product A, Product B, and Product C.

Observation

Product C has the highest available quantity of 300 units, while Product B has the lowest available quantity of 175 units.

2. Stacked Bar Chart

Aim

To generate a stacked bar chart showing the quantity of each product within different product categories.

R Code

product_name <- c("Product A", "Product B", "Product C")

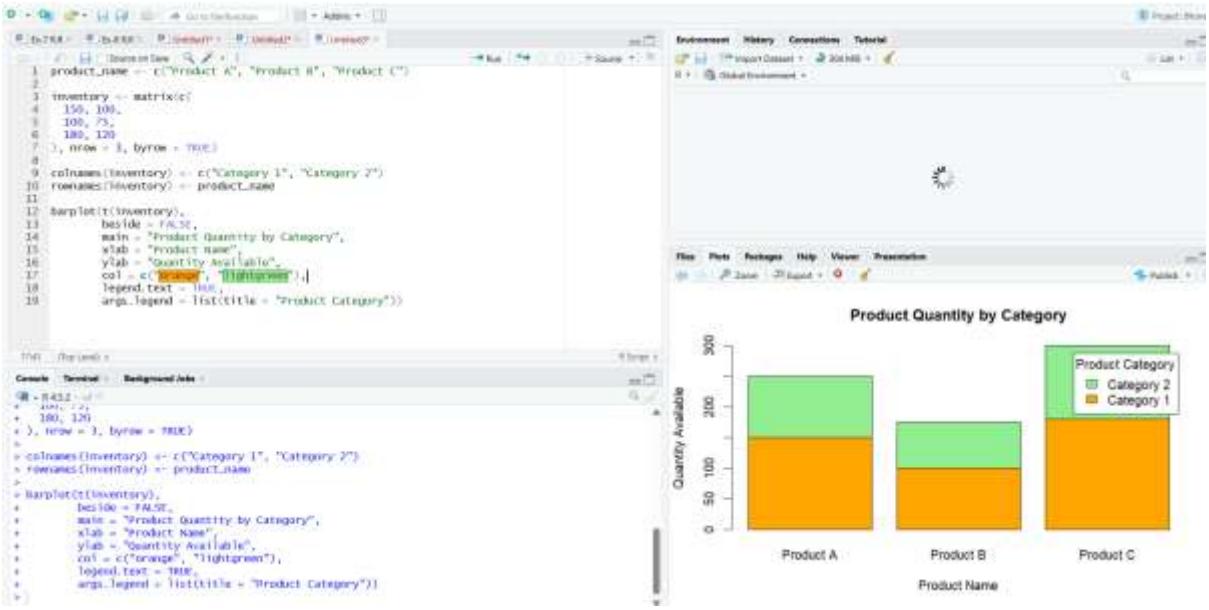
inventory <- matrix(c(

150, 100,

100, 75,

```
    180, 120
), nrow = 3, byrow = TRUE)
colnames(inventory) <- c("Category 1", "Category 2")
rownames(inventory) <- product_name
barplot(t(inventory),
        beside = FALSE,
        main = "Product Quantity by Category",
        xlab = "Product Name",
        ylab = "Quantity Available",
        col = c("orange", "lightgreen"),
        legend.text = TRUE,
        args.legend = list(title = "Product Category"))
```

Output



A stacked bar chart showing the total quantity of each product divided into two illustrative categories.

Observation

Product C has the highest total quantity of 300 units, followed by Product A with 250 units and Product B with 175 units. The category-wise quantities in this example are assumed for demonstration.

3. Product Inventory Table

Aim

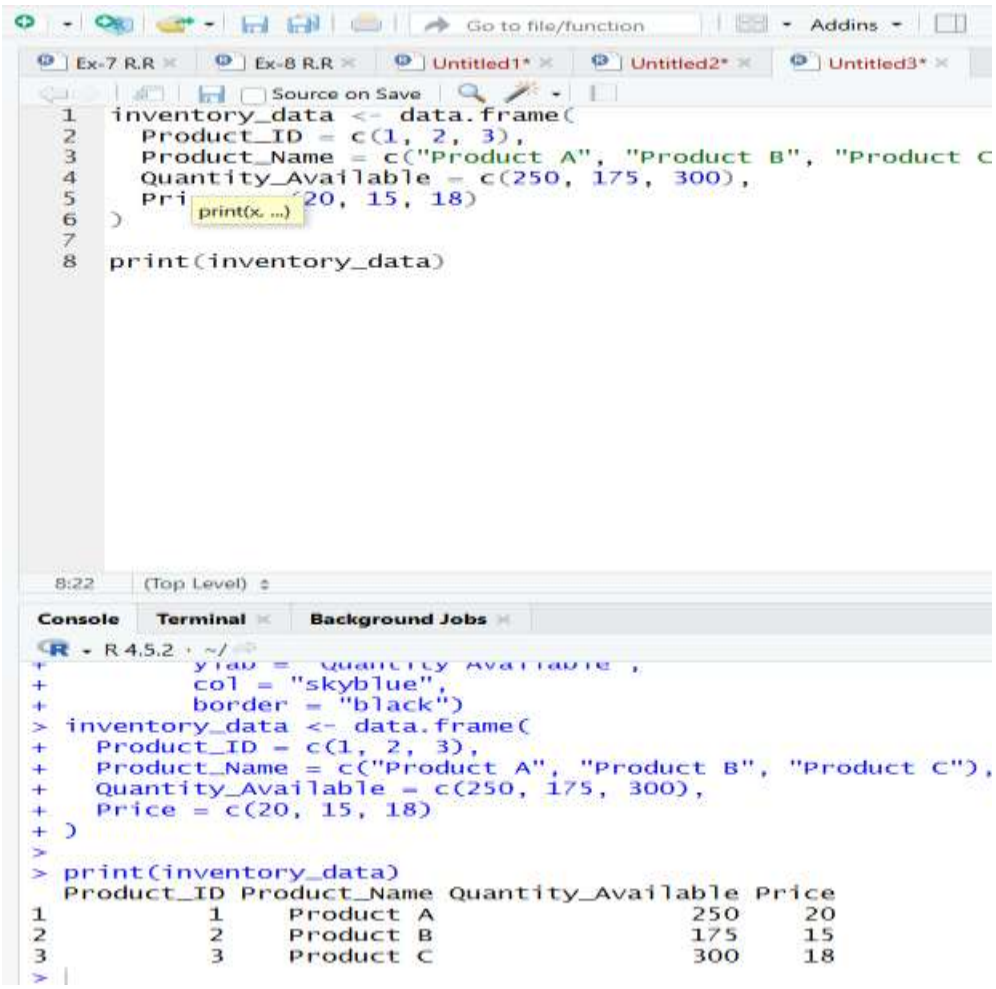
To create a table displaying the inventory data for each product, including product ID, product name, quantity available, and price.

R Code

```
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

print(inventory_data)
```

Output



Observation

The table clearly displays the inventory details of all three products. Product C has the highest quantity, while Product B has the lowest price of $15.

4. Tableau Dashboard

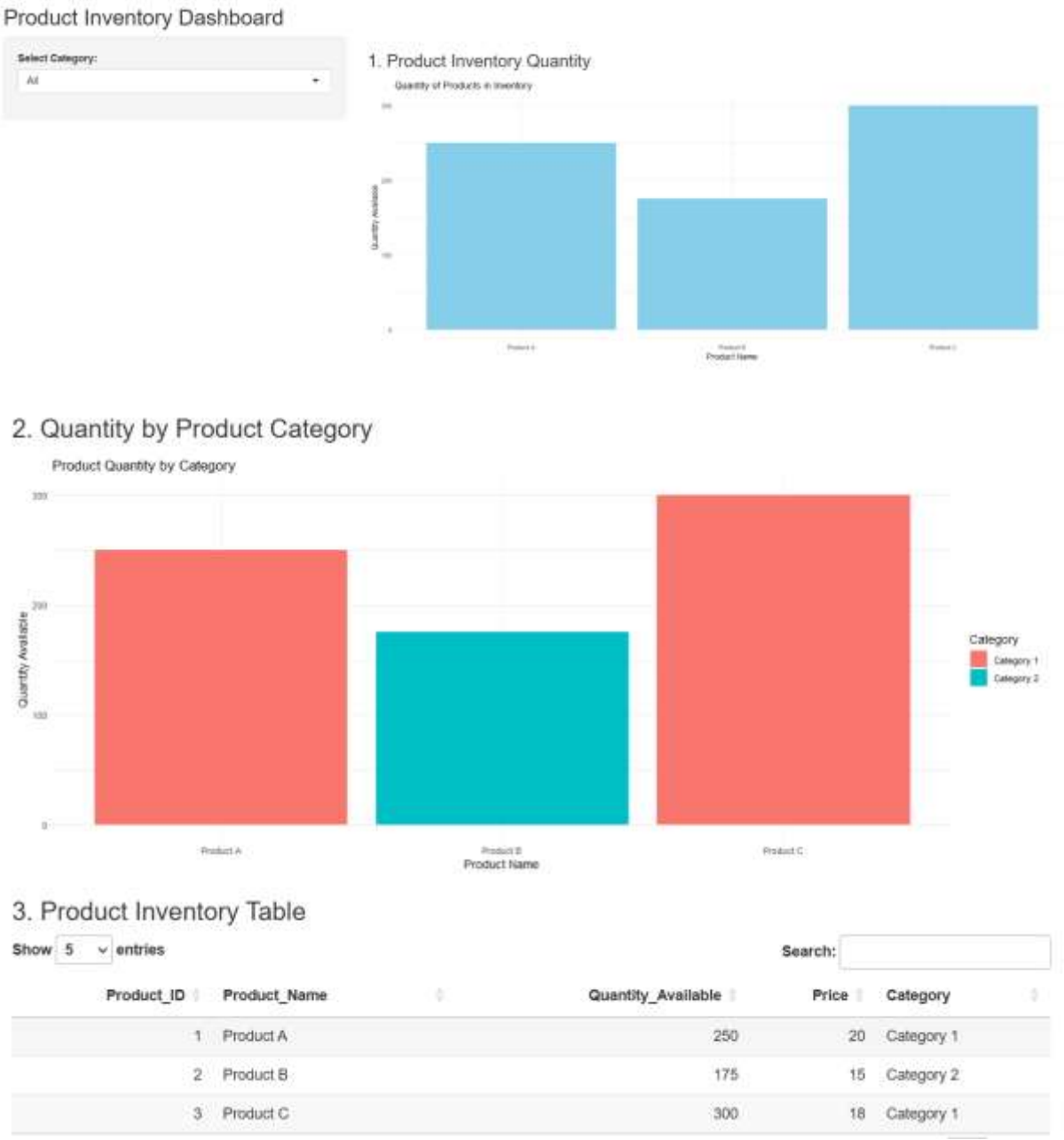
Aim

To develop an interactive dashboard combining the bar chart, stacked bar chart, and product inventory table for exploring inventory data.

Tableau Procedure

1. Open Tableau.
2. Import the Product Inventory dataset.
3. Create a bar chart using Product Name and Quantity Available.
4. Add a suitable chart title and axis labels.
5. Create a stacked bar chart using Product Name, Quantity Available, and Product Category.
6. Assign different colors to the product categories.
7. Create a table containing Product ID, Product Name, Quantity Available, and Price.
8. Select Dashboard → New Dashboard.
9. Drag the inventory bar chart into the dashboard.
10. Drag the stacked bar chart into the dashboard.
11. Drag the product inventory table into the dashboard.
12. Arrange the charts and table neatly.
13. Add suitable titles, labels, and filters for interactive exploration.
14. Save the dashboard.

Output



An interactive dashboard displaying product quantities, category-wise inventory quantities, and the inventory data table.

Result

The product inventory dataset was successfully analyzed using a bar chart, stacked bar chart, and table in R. A dashboard can combine these visualizations to provide an interactive view of product quantities, categories, and prices.